

GEOGRAPHICAL INFORMATION SYSTEM (GIS) APPLICATION ON A DESCRIPTIVE SOIL COMPOSITION ANALYSIS IN NASSAU COUNTY, NEW YORK

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Abstract

In respect to previous knowledge of the Wisconsin Glacial Episode, our research focused on analyzing the current soil compositional make-up of Nassau County, New York in contrast to the soil compositional make-up dating back to the early 1900's. This research aims to construct a descriptive soil analysis while comparing how the soil of Nassau County has changed over time while taking into account various environmental factors. Those factors being climate change and erosion. Soil survey data provided by Field Operations of the Bureau of Soils was used in conjunction with 2020 Soil Survey Data collected by the United States Department of Agriculture (USDA), Natural Resources Conservation Service (NRCS). Two bar graphs were then created to compare the percentages of sand, silt and clay for each soil type. Those numbers were then used to determine a trend which showed that the soil types located in Nassau County have slowly evolved over time.

Introduction

Long Island topography is a direct product of the Wisconsin Glacial Episode that occurred approximately 22,000 years ago leaving behind two of the most well-known topographic features of Long Island. Both of these bi-products consist of glacial till which can be classified as Acrisol, which is made-up of boulders, gravel, sand, silt and clay which can be traced back to the movement and initial melting of the continental ice sheets. (Monti, 1968) During the early 1900s Nassau County consisted of five primarily soil types. Hempstead Loam which contained high percentages of soil, Hempstead Gravelly Loam which contained high percentages of silt, Galveston Sandy Loam which contained high percentages of sand and Galveston Clay that was primarily composed of mud and gravel.

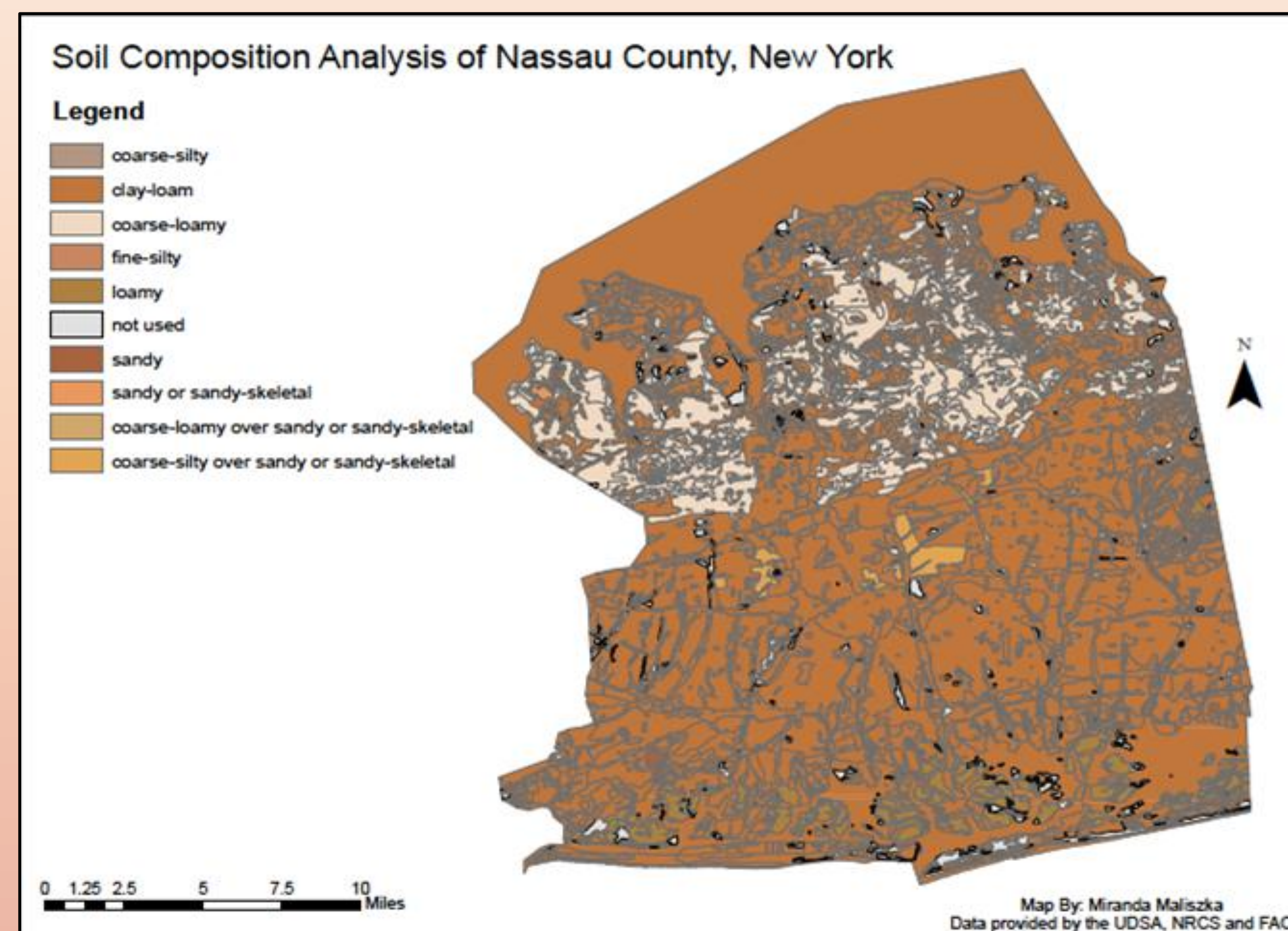


Figure 1: Map created by this research showcasing the soil composition of Nassau County, New York using data provided by the United States Department of Agriculture, Natural Resources Conservation Service

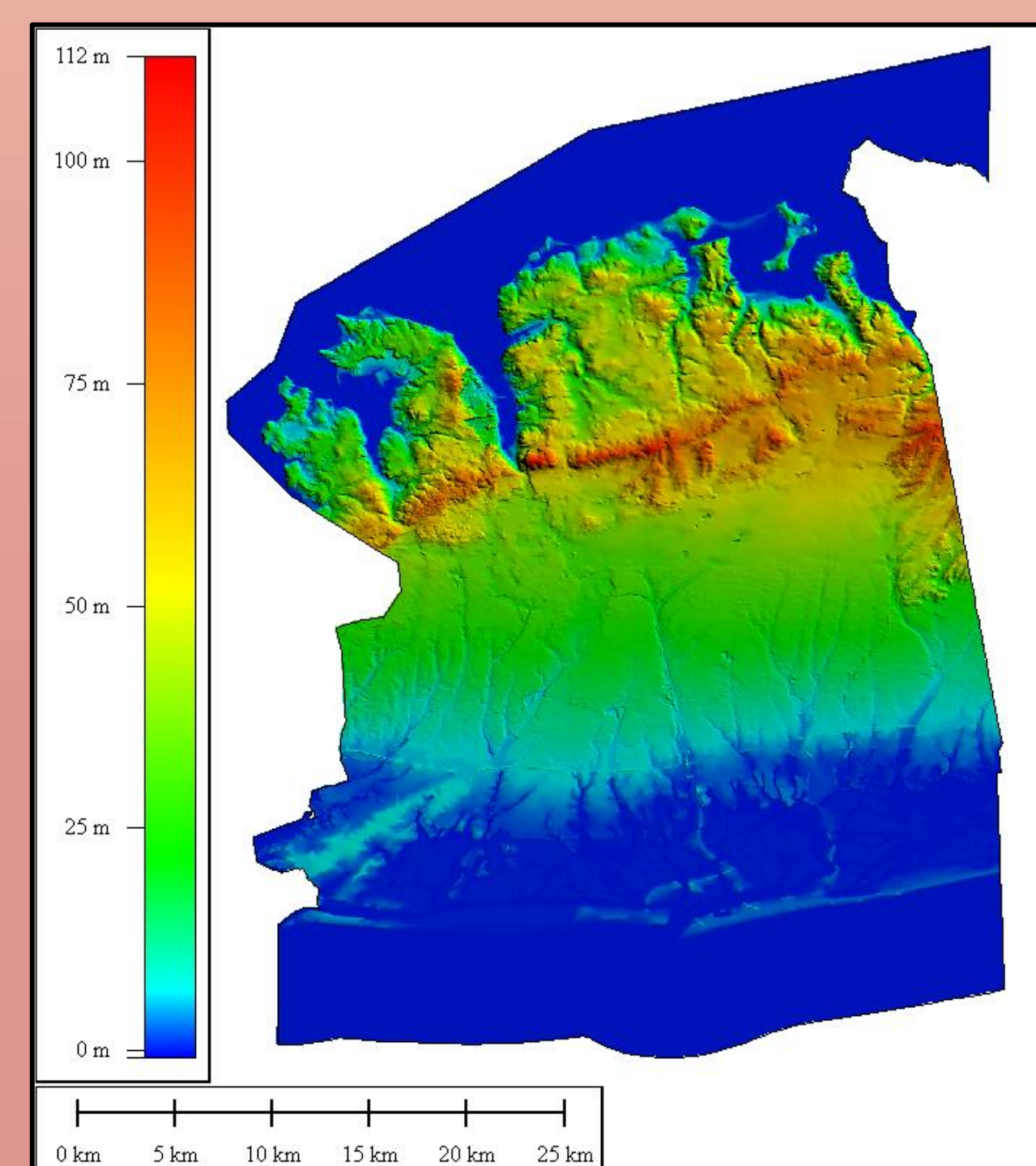


Figure 2: Digital Elevation Model (DEM) of Nassau county in Long Island, NY created in Global Mapper using a 10-meters resolution from the National Elevation Dataset (NED) shows the current topography.

Methods

- Data provided by the USDA, NRCS was used to compose a Soil Composition Analysis of Nassau County, NY. The SSURGO database was used to run the Nassau County tabular data which generated hundreds of samples while ArcMap was then used to process and project the spatial data which excluded soil text classification as one of the attributes.
- Unsurveyed areas were identified using a Digital Soil Map of the World provided by the FAO. All of the areas that were not surveyed were identified with the soil text classification of Dystric Cambisols (Bd) with the following compositional make-up: 32.7% sand, 30.3% silt and 37.1% clay.
- The soil triangle was used to identify the following soil composition which is clay-loam. The data was then reclassified as Clay-Loam. To provide a visual representation of both the 1903 and 2020 soil survey data, bar graphs were constructed.

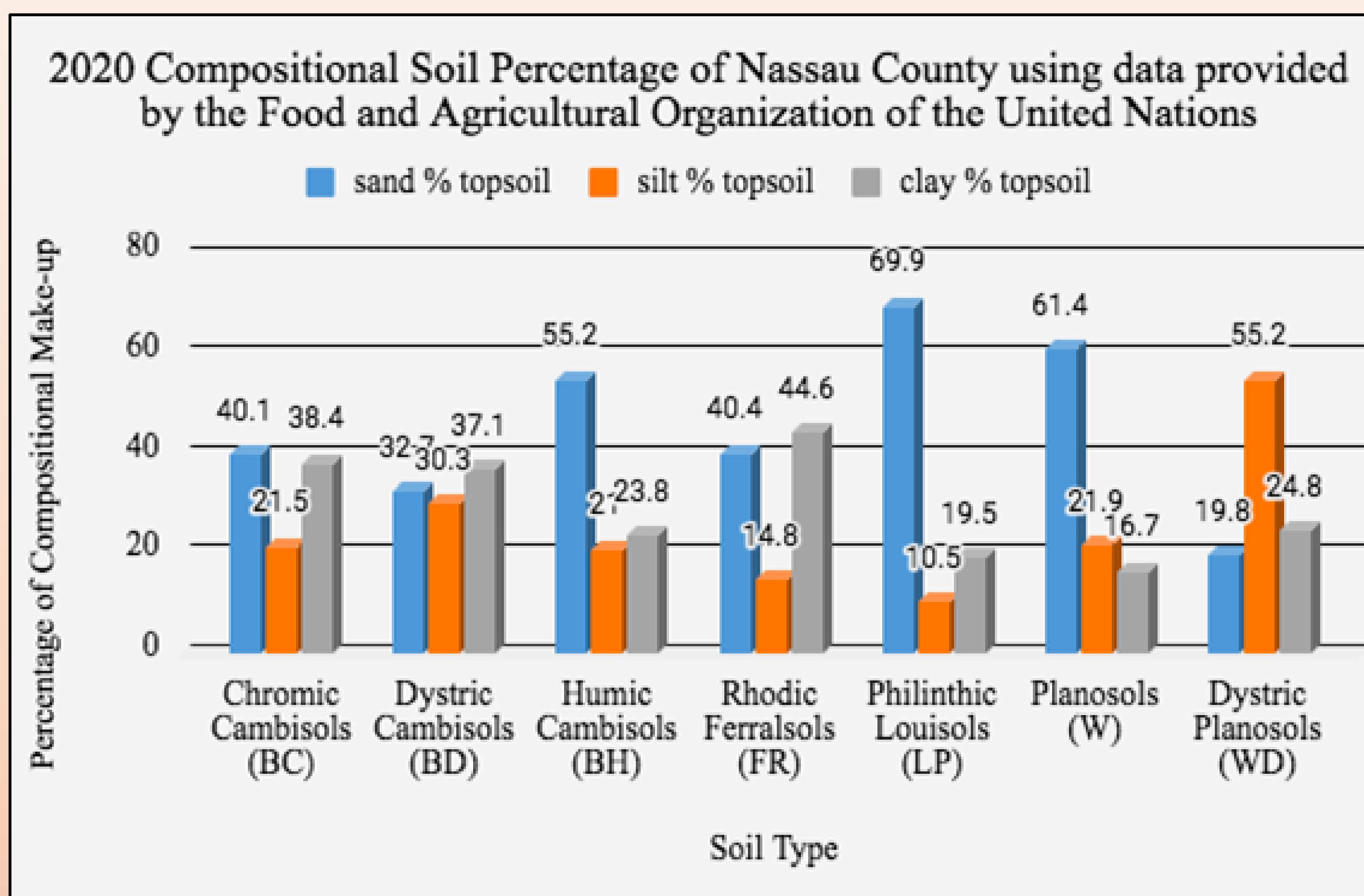


Figure 3: Bar Graph representing the topsoil percentage breakdown of sand, silt and loam in order to identify the most common soils in Nassau County, New York by using data provided by the Food and Agricultural Organization of the United Nations (2020).

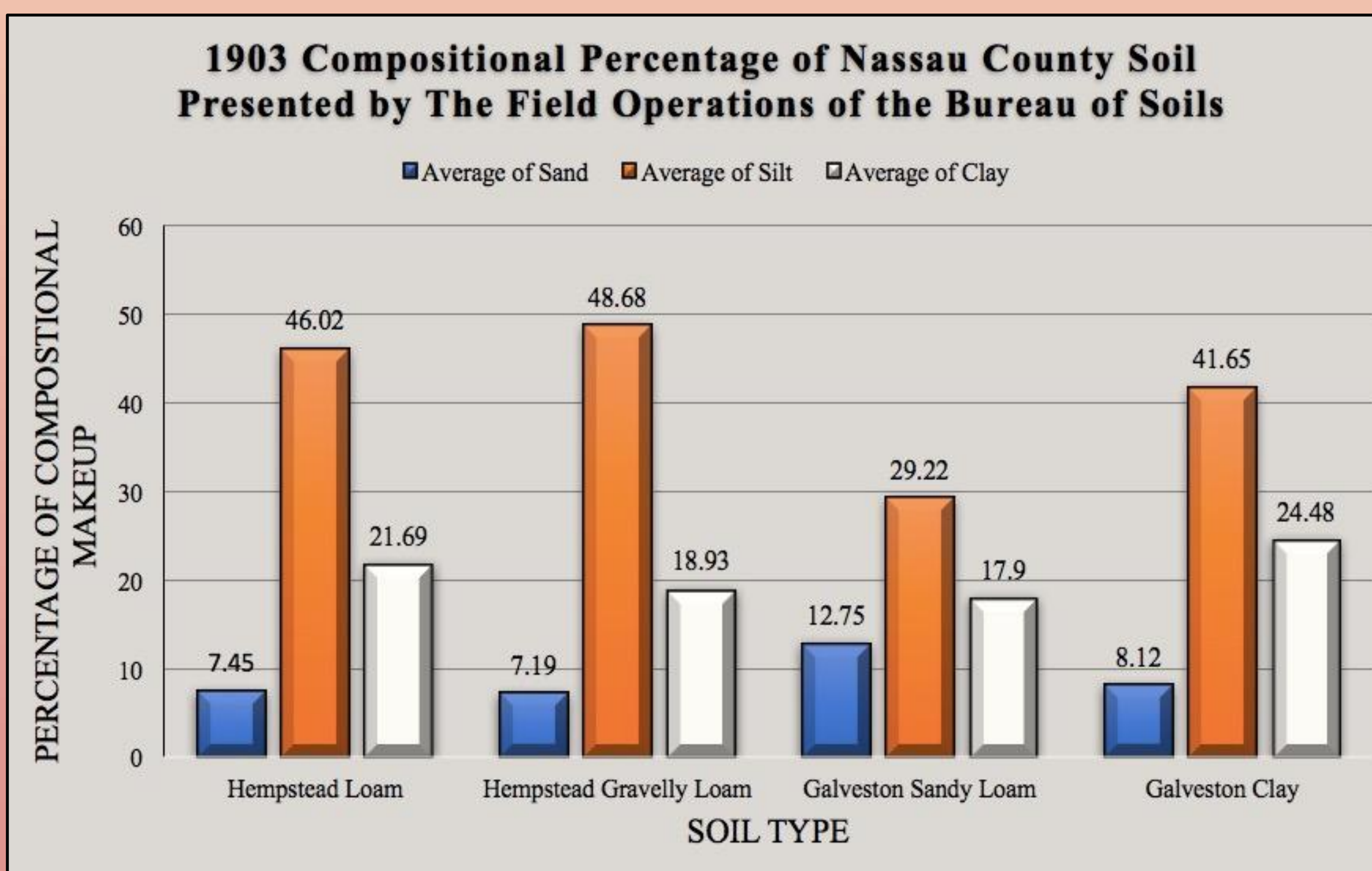


Figure 4: Bar Graph representing the soil make-up percentages of sand, silt and clay using data provided by The Field Operations of the Bureau of Soils (1903).

Results

- $\frac{3}{4}$ of the most common soil types in Nassau County dominate more than 50% of their compositional make-up with sand (Figure 3), meaning that Philinthic Louisols (69.9%) and Planosols (61.4%) can be classified as Sandy Loam
- Humic Cambisols (55.2%) have a high percentage of sand, classifying it as Sandy Loam Clay
- The 1903 soil survey data shows a strong correlation between the percentage of silt in all four of the sample (Figure 4)
- The highest percentage of silt can be found in Hempstead Gravelly Loam (48.6%), while the lowest percentage of silt can be found in Galveston Sandy Loam (29.22%)

Discussion

- 1903 survey data, strong dominance of silt between all four samples (Fg 3.), while today sand is more prominent amongst all seven samples (Fg 4.)
- As a result of climate change, there has been a slow increase in precipitation rate, which has caused greater amounts of chemical and mechanical weathering
- In the United States, the amount of precipitation has been increasing at a rate of 0.17 inches per decade (EPA, 2016)
- Comparing today's soil composition to 1903, there is more sand present

Conclusion

Using data from the USDA, NRCS and FAO of the United Nations, a map was produced to visually represent the current soil types located in Nassau County, New York. In addition, 1903 soil survey data provided by the Field Operations of the Bureau of Soils was used further to compare the progression of soil over the past century. With that knowledge in conjunction with current environmental factors such as climate change and erosion, a better understanding of how the soil composition has changed over time was cultivated. In conclusion, over the past hundred and fifteen years a county once dominating in silt has slowly been transformed to a ground of higher percentages of sand and clay.

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